

KARKI SENTHILKUMAR

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AI & AUTOMATION ENGINEER | FULL-STACK DEVELOPER

PROFESSIONAL SUMMARY

Results-driven professional transitioned to Software Engineering, specializing in AI-driven quantitative systems, seamless API integration, and full-stack development. Proven architect of the Zenith ecosystem—a high-performance ML prediction engine coupled with React-based visualizations and exhaustive n8n automation workflows. Blends a background in complex process management with cutting-edge tech (Python, FastAPI, TypeScript, React 18) to deliver robust, high-availability data solutions and automation tools.

TECHNICAL SKILLS

Programming Languages: Python 3.12, TypeScript, JavaScript, HTML/CSS, Node.js

AI & Machine Learning: XGBoost, Pandas, Scikit-learn, Feature Engineering, Explainable AI (XAI)

Backend & Automation: FastAPI, n8n workflow orchestration, REST APIs, WebSockets

Frontend Development: React 18, Vite, TailwindCSS, Framer Motion, Recharts

Trading / Data Systems: Angel One API, Google Sheets API, Real-time Data Normalization, Algorithmic Execution

FEATURED PROJECTS

Zenith Intelligence Engine (AI Trading AI) | Python, FastAPI, XGBoost, Pandas

- Engineered a 57-feature ML prediction pipeline fetching real-time market data matching signals at 14-52ms latencies directly from Angel One live data.
- Implemented a dual-brain safety architecture using Python singleton patterns.
- Integrated Options Chain intelligence detecting Smart Money Concepts (SMC) breakouts.

Zenith Advanced Trading Terminal | React 18, TypeScript, Vite, TailwindCSS

- Built an 11-page command center frontend providing intuitive XAI (Explainable AI) transparency.
- Constructed live dynamic signal feed monitoring with sub-100ms render latency and Glassmorphism aesthetics.
- Integrated a backtest engine UI and engine diagnostic components to interface natively with FastAPI.

Complete Intraday Options n8n Bot / Flow Platform | n8n, Node.js, REST API, Google Sheets

- Designed a 24-node fully automated execution workflow that queries FastAPI models and places market entry, stop-loss, and target exit orders.
- Implemented background exit order monitors (CRON jobs) automatically reconciling state against live ledger active orders.
- Achieved comprehensive P&L logging natively inside Google Sheets seamlessly using event-driven architectures.

PROFESSIONAL EXPERIENCE

Senior Process Executive | Cognizant Technology Solutions (Jul 2022 - Present)

- Developed strong analytical capabilities resolving complex workflow disputes and managing data operations accurately.

Senior Process Executive | Sutherland Global Services (Nov 2019 - Jul 2022)

- Leveraged critical problem-solving skills inside data-heavy environments.

Site Admin | SEC (P) LTD., (May 2019 - Nov 2019)

EDUCATION

B.Sc. Information Technology

Shri Nehru Mah Vidhyala College of Arts & Science, Coimbatore, Tamil Nadu

Graduated: 2019